

Logical Agent Case Studies – Problem Statement

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. Knowledge base: (I) If a patient has Fever AND Cough $\rightarrow$ Possible Flu. (II) If a patient has Fever AND Rash $\rightarrow$ Possible Measles. (III) If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia. (IV) Patient A: Fever=True, Cough=True, Breathlessness=True.

(a) Represent using Propositional Logic and define symbols. (b) Apply Forward Chaining and show every inference. (c) Prove Patient A has Pneumonia using Backward Chaining and draw the goal reduction tree.

Case Study 2 — Autonomous Traffic Management Agent

FOL KB: (I) $\forall x \text{ Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$. (II) $\forall x \text{ ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$. (III) $\forall x \forall y \text{ Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$. Facts: $\text{Vehicle}(\text{Ambulance})$, $\text{EmergencyType}(\text{Ambulance})$, $\text{Vehicle}(\text{CarA})$, $\text{Behind}(\text{CarA}, \text{Ambulance})$.

(a) Apply Unification and show the MGU. (b) Use Resolution to prove $\text{Proceed}(\text{CarA})$. (c) Analyze whether the KB handles two simultaneous emergency vehicles.

Case Study 3 — Agricultural Expert Reasoning System

P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$. P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$. P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$. Facts: $\text{SoilDry}=\text{True}$, $\text{CropWheat}=\text{True}$.

(a) Convert all sentences to CNF. (b) Prove ApplyDripMethod by Resolution Refutation and derive $\blacksquare$. (c) Remove SoilDry and analyze whether CropAtRisk still follows.

Case Study 4 — Wumpus World Knowledge Agent

Percepts: $\text{Stench}[1,2]$, $\text{Breeze}[1,1]$, $\text{Glitter}[2,2]$. Rules: $\text{Stench}[1,2] \rightarrow \text{WumpusAdjacent}[1,2]$; $\text{Breeze}[1,1] \rightarrow \text{PitAdjacent}[1,1]$; $\text{Glitter}[x,y] \rightarrow \text{Gold}[x,y]$; $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$.

(a) Construct the belief state using Modus Ponens. (b) Use Forward Chaining to identify possible Wumpus and Pit locations. (c) Evaluate whether $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is safe.